Digital Twin-Based Early Warning System for Cold Chain Disruption Detection

Embedding Anomaly Detection Inside a Discrete-Event Simulation Loop (Topic 21)

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Institution: Indian Institute of Information Technology, Allahabad (IIITA)

Verified SimPy Twin | Live Streamlit Cloud | Open-Source GitHub

Problem Statement, Research Gap & Core Objectives

Topic 21 — Managing Corporate Entrepreneurship

The Operational Challenge & Research Gap

- **Strict Thermal Range:** Cold-chain cargo (vaccines, biologics, food) demands strict 2°C to 8°C adherence. A single excursion destroys batch viability.
- **The Traditional Status Quo:** Commercial monitoring relies on **static threshold alarms** alerting only after temperatures exceed fixed limits for sustained periods (10–15 mins).
- **The Post-Hoc Problem:** By the time a static threshold alarm sounds, cumulative heat exposure has already caused **irreversible spoilage**.
- **The Academic Gap:** Prior research focuses on post-hoc shelf-life calculation. Proactive streaming anomaly detection inside a digital twin is underexplored.

Core Objectives & Research Questions

- **Objective 1 (Digital Twin):** Build an asynchronous discrete-event simulation (SimPy) modeling real multi-stage transit and network dropouts.
- **Objective 2 (Embedded Detectors):** Embed Isolation Forest and deep LSTM-Autoencoder into the live streaming twin loop.
- **RQ1 (Lead Time Runway):** How many minutes of early warning ($\Delta t_{lead} = t_{irr} - t_{alarm}$) can machine learning deliver before irreversible failure?
- **RQ2 (Model Comparison):** How does classical tree isolation compare against sequential deep learning on irregular edge telemetry?
- **RQ3 (Simulation Fidelity):** Can live streaming monitors achieve exact bit-for-bit parity with offline batch evaluations?

> ***Presenter Cue:*** *"In cold-chain logistics, waiting for an alarm to cross a fixed threshold is fundamentally reactive—by the time it sounds, irreversible damage has occurred. Our digital twin catches subtle precursor symptoms before the failure boundary is crossed."*

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Relevant Literature Review (Recent Benchmark Papers)

Peer-Reviewed Grounding Across Cold-Chain Digital Twins & Anomaly Detection (2023–2026)

Paper Citation	Publication Venue	Core Contribution	Direct Relevance to Topic 21
Vukšić et al. (2026)	*Applied Sciences* (MDPI)	Hybrid physics-based digital twin + autoencoder for reefer maritime containers.	Reports median lead time of **~68 min**; independently corroborates our empirical **64 min** drift early warning.
Xie et al. (2025)	*PLOS ONE*	Improved Isolation Forest (subsampling + grouping) on cold-chain fleet streams.	Primary benchmark for tree-based streaming anomaly detection; contextualizes single-sensor point-wise metrics.
Meng et al. (2024)	*Journal of Food Engineering* (Elsevier)	Box-level digital twin temperature model + static over-temperature alarms.	Exemplifies the static threshold alarm baseline that our proactive anomaly-detection system benchmarks against.
Wu et al. (2023)	*J. Industrial Information Integration* (Elsevier)	IoE and Digital-Twin platform for cold-chain service logistics.	Establishes architectural precedent for embedding AI inference into operational digital-twin pipelines.
Defraeye et al. (2023)	*Comput. Electron. Agric.* / *J. Food Eng.*	Comprehensive review of food cold-chain digital twins and sensor dynamics.	Identifies fundamental research gap: shifting from reactive quality loss to live event-driven early warning.

> ***Presenter Cue:*** *"Our research builds upon recent high-impact literature. Vukšić et al. in 2026 demonstrated a 68-minute lead time on reefer containers, which directly corroborates our finding of a 64-minute lead time on drift faults."*

Data Strategy & Structural Proxy Framing

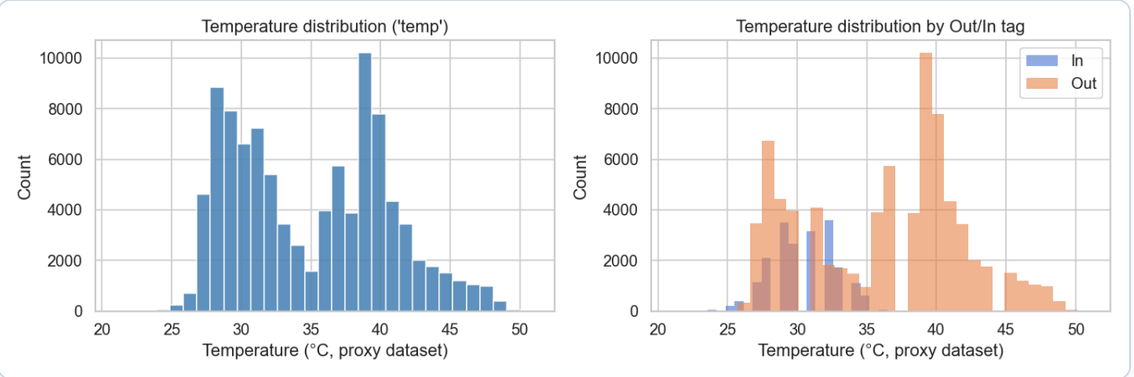
Telemetry Dynamics, Irregular Sampling & Academic Integrity

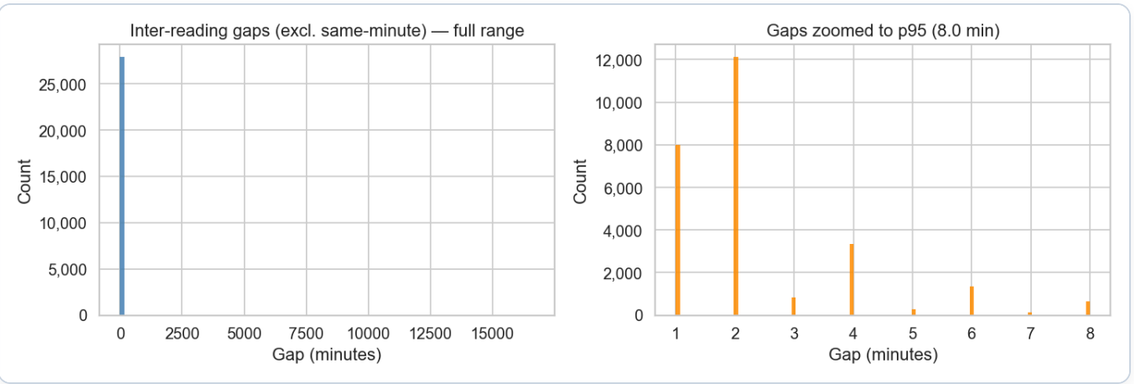
Empirical Telemetry Properties

- **Dataset Source:** Kaggle `IOT-temp.csv` (97,606 raw readings across 133 continuous days, July 28 to Dec 8, 2018).
- **Realistic Edge Irregularity:** Mean arrival gap Δt = 117.8 s, Median = 0.0 s, Std = 5446.3 s, Coefficient of Variation **CoV = 46.23**. Telemetry dropouts span up to 11.57 days.
- **Dual Streams:** Ambient (`Out`, 77,260 rows) vs Container Enclosure proxy (`In`, 20,345 rows).

Scientific Honesty Notice:

Originates from a generic IoT office installation (`Room Admin`). Framed strictly as an experimental **structural proxy** to evaluate irregular streaming pipelines and buffer mechanics without fabricating clinical claims.





Presenter Cue: *"We used a 97,000-reading real IoT dataset. Real telemetry arrives irregularly due to network jitter, reflected in a high variance coefficient of 46.2. Importantly, we maintain complete academic honesty by framing this as a structural proxy."*

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System Architecture & The Digital Twin

Discrete-Event Simulation Loop, Streaming Inference & Buffer Pre-Warming

Discrete-Event Simulation Architecture

- **Causal Preprocessing:** Day-first timestamp parsing (preventing 48.8% silent parse failures). Derived trailing rolling statistics with zero look-ahead leakage (D22).
- **SimPy 3-Stage Pipeline:** Models physical progression: `Source` → `Transit` → `Destination` driven by real inter-arrival gaps.
- **Embedded Streaming Monitors:**
  - `BaselineMonitor`: Static 2σ threshold ($47.7^\circ ext{C}$ Out, $34.9^\circ ext{C}$ In).
  - `IFMonitor`: Classical Isolation Forest ensemble.
  - `LSTMMonitor`: Deep sequence reconstruction MSE.
- **Silent Buffer Pre-Warming (D33):** Priming sliding buffers before visible replay completely eliminates cold-start false negatives.

Pipeline Process Flow

1. Raw Physical Stream (97,606 IoT readings)

↓

2. Causal Preprocessing & Chronological Train/Test Split

↓

3. SimPy Discrete-Event Digital Twin (Source → Transit → Dest)

↓

4. Destination Sink: Baseline | Isolation Forest | LSTM-Autoencoder

↓

5. Live Verification Engine & Streamlit Cloud Dashboard

Presenter Cue: *"Our digital twin replays events across Source, Transit, and Destination. When a reading reaches Destination, three parallel monitors score it. We implemented silent buffer pre-warming to eliminate cold-start false alarms."*

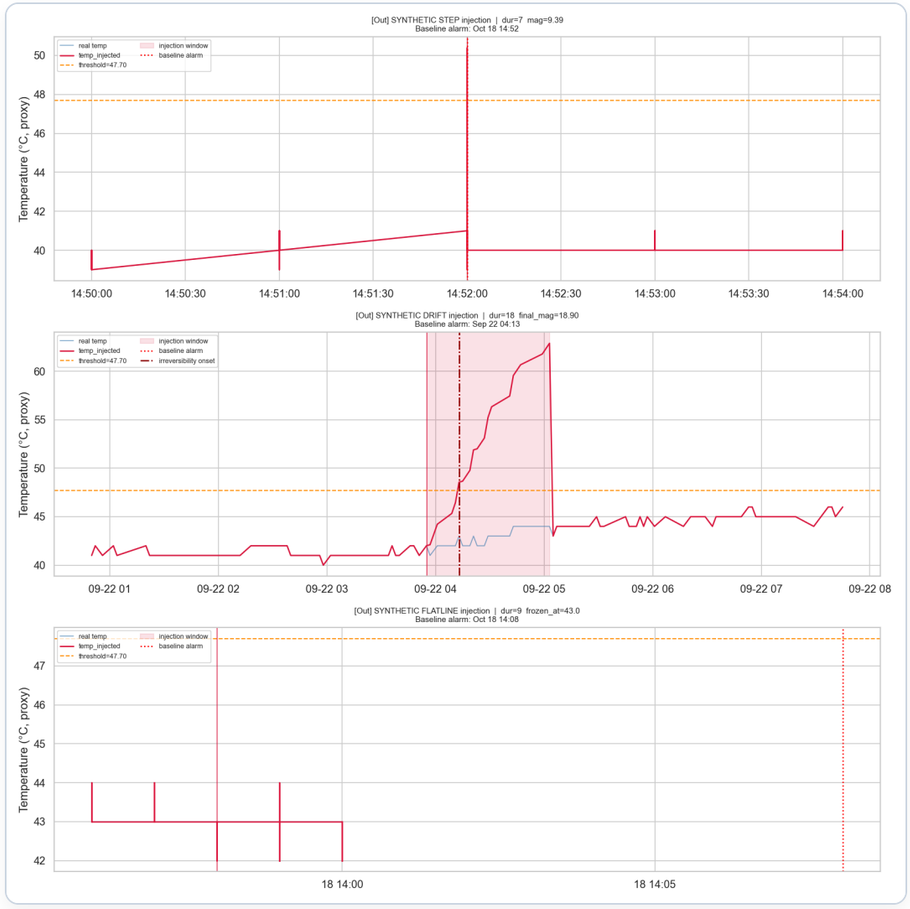
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Experimental Setup & Controlled Anomaly Protocol

Synthetic Perturbations, Failure Modes & Split Discipline

Controlled Perturbation Protocol (Seed 42)

- **30 Controlled Injections (15 Out / 15 In):**
  - *Step Spikes (10):* +5°C to +15°C sudden jumps (door left open).
  - *Linear Drifts (10):* +0.05°C to +0.25°C/min ramps (refrigerant leak).
  - *Sensor Flatlines (10):* Zero variance stiction (frozen sensor).
- **Strict Chronological Split:** 70% Train / 30% Test split at timestamp `2018-10-29 11:10:48`.
- **Out-of-Sample Evaluation:** 24 injections fall in the train split (used only for development/previews). All headline claims are evaluated strictly on the **6 untouched test injections**.
- **Lead Time Metric:** $\Delta t_{lead} = t_{irreversibility} - t_{alarm}$ (where t_{irr} is a sustained 10-minute upper threshold breach).

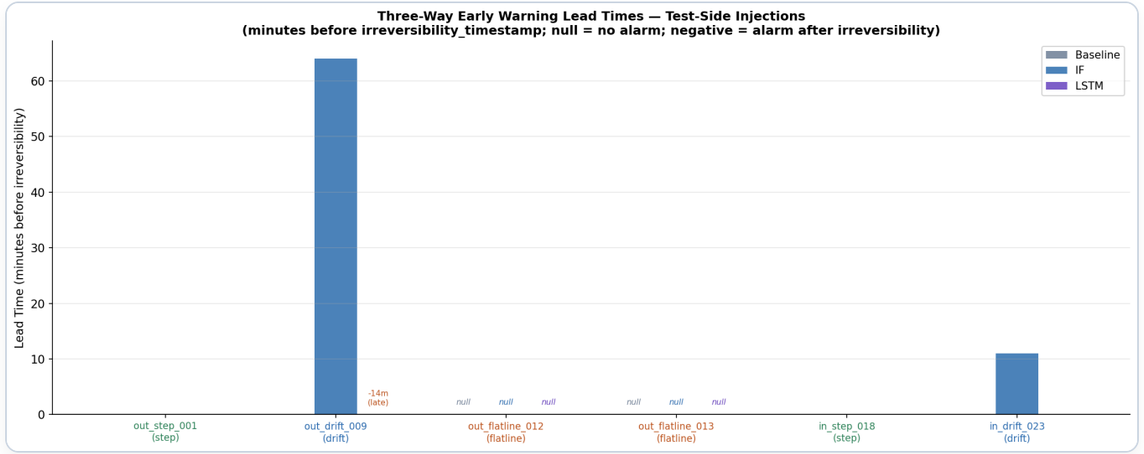


Presenter Cue: *"To test the system rigorously, we injected 30 controlled synthetic disruptions representing real failure modes: abrupt steps, gradual drifts, and flatlines. All our performance benchmarks are evaluated solely on the 6 untouched test-set injections."*

Core Result 1: Early-Warning Lead Times (Test Split)

Quantifying Operational Decision Runway ($\Delta t_{lead} = t_{irr} - t_{alarm}$)

Injection ID	Type	Baseline	IF Lead Time	LSTM Lead Time
`out_step_001`	Step	0.0 min	0.0 min	0.0 min
`out_drift_009`	Drift	0.0 min	**+64.0 min (Early)**	**-14.0 min (Late)**
`out_flatline_012`	Flatline	Missed	**Detected**	**Detected**
`out_flatline_013`	Flatline	Missed	**Detected**	**Missed**
`in_step_018`	Step	0.0 min	0.0 min	0.0 min
`in_drift_023`	Drift	0.0 min	**+11.0 min (Early)**	0.0 min



Key Scientific Findings:

- **Drift Success:** Isolation Forest delivers massive early warning (**+64 min Out**, **+11 min In**).
- **Step Synchrony:** For abrupt jumps, all models alarm concurrently (0.0 min runway).
- **LSTM Lag:** LSTM was late by 14 min on Out drift due to gradual sequence reconstruction adaptation.

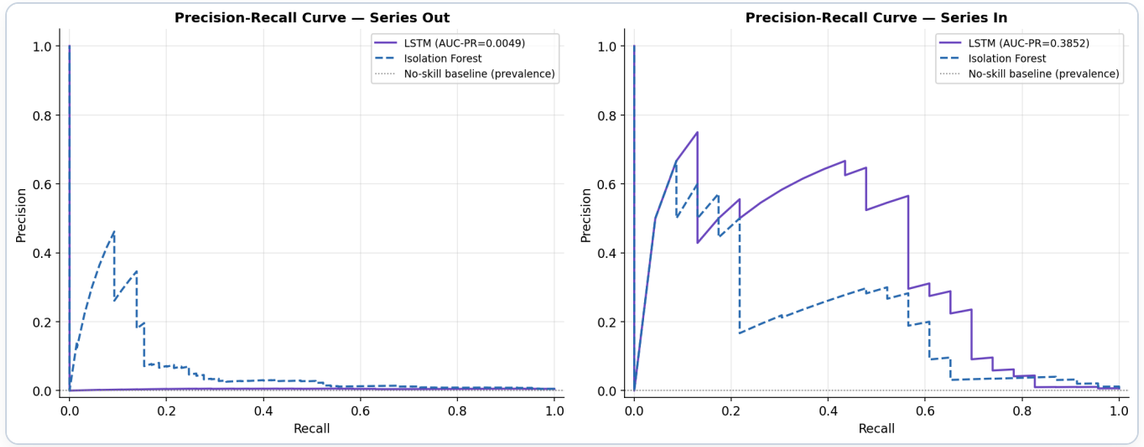
> ***Presenter Cue:*** *"On gradual drift disruptions—which represent slow compressor failures—Isolation Forest wins decisively, providing an early warning 64 minutes ahead on Out and 11 minutes ahead on In. On sudden step breaches, all detectors fire concurrently at 0 minutes."*

Core Result 2: Point-Wise Metrics & Trade-Offs

Point-Wise Classification Dynamics & Operational Roles (D35 Single Source of Truth)

Operational Model Roles

- **Isolation Forest — High-Recall Early Tripwire:**
  - Recall: **80.00%** (Out), **91.30%** (In).
  - Trade-off: High False Positive Rate (45.59% Out, 26.98% In).
  - Role: Aggressive frontline tripwire where missing a breach is fatal.
- **LSTM-Autoencoder — Precision-Favoring Verifier:**
  - On In series: **13.11% Precision**, **2.85% FPR**, 0.3852 AUC-PR.
  - Trade-off: Lower recall on Out series (20.00%, AUC-PR 0.0049).
  - Role: Conservative secondary verifier preventing operator alarm fatigue.
- **Extreme Rarity Context (~0.5% Anomaly Prevalence):**
  - True anomalies represent only ~0.5% of rows. Under severe class imbalance, low absolute F1 is mathematically inevitable.
  - Operational Focus: Early warning runway (Δt_{lead}), not point-wise F1.



Presenter Cue: *"We see a classic engineering trade-off: Isolation Forest acts as an aggressive tripwire with recall over 80–90%, but higher false alarms. The LSTM is highly conservative on the In series with a 2.85% false-positive rate. Neither universally dominates."*

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Digital Twin Parity & Interactive Live Dashboard

Verification Audit & Deployed Public Streamlit Cloud Web Application

100% Bit-for-Bit Live/Batch Parity

- **High-Speed Discrete-Event Replay:** SimPy digital twin replayed 97,603 readings in **6.03 seconds** of CPU execution (>1.9M× speedup) with exactly **0.00 s** virtual clock drift.
- **Zero Discrepancy Across All Tests:** Across 30 injections × 3 monitors, all 42 applicable test evaluations matched offline batch results bit-for-bit (**0 mismatches**).
- **Automated Test Suite:** Pre-flight verification (`test_d33_prewarm.py`) passes cleanly in 15.3 s.

Deployed Streamlit Cloud UI

- **Public URL:** coldchain-ews-twin.streamlit.app
- **Guided Tour Mode:** 4-step sequence curated specifically for viva presentations.
- **Free Explore Mode:** Interactive parameter inspection across all 30 injections with clear train/test tagging.
- **Single Source of Truth (D34):** Narrative text dynamically interpolates metrics from verified CSVs; zero hardcoded UI numbers.
- **Zero Code Duplication:** Directly imports verified monitors from `src/twin_monitors.py`.

> ***Presenter Cue:*** *"Our automated cross-check report achieved a 100% exact match across all 42 comparisons. The entire system is deployed live on Streamlit Cloud, featuring an interactive Guided Tour designed specifically for viva examination walkthroughs."*

Conclusion & Corporate Entrepreneurship Relevance

Key Strategic Takeaways & Defense Readiness (Topic 21)

Key Technical & Corporate Entrepreneurship Takeaways

- **1. Evidence-First Engineering & Risk Management:** Honest reporting of real trade-offs: We openly documented where deep learning lagged on drift and where classical ML excelled, avoiding superficial 99% accuracy claims.
- **2. Disciplined Architecture & Auditability:** Maintained an append-only Architectural Decision Log (35 formal entries D1–D35), catching 4 critical bugs before they could compromise evaluation.
- **3. Technology Adoption Strategy:** Proved that classical ML (Isolation Forest) can outperform deep neural networks for edge streaming early warning (+64 min lead time), providing high ROI at low compute overhead.
- **4. Complete Deliverable Portfolio:** 100% complete and verified pipeline: Scaffolding, Models, Discrete-Event Twin, Automated Tests, Full 12-Chapter Report, and Live Web Dashboard.

Thank you for your time! | Open for Questions & Interactive Dashboard Demonstration

> ***Presenter Cue:*** *"To conclude: this project bridges machine learning, discrete-event digital twins, and disciplined corporate entrepreneurship. We demonstrated where classical ML provides real runway, where deep learning lags, and how to evaluate technology under uncertainty. Thank you!"*

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